Ex 3a: Model Planning and Building – Classification

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence and Data Science
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Ex3a: Model Planning and Building - Classification

Learning Objective:
The objective of this experiment is to enable students to plan, build, and evaluate basic machine learning models using real-world datasets.

Problem Statement:
A telecommunications company has noticed that a growing number of customers are discontinuing its services, resulting in reduced revenue and increased customer acquisition costs. After completing data preprocessing and exploratory analysis, the company now aims to develop predictive models that can identify customers who are likely to churn. Early prediction of customer churn will enable the company to implement targeted retention strategies, improve customer satisfaction, and reduce business losses.

As a Data Scientist, your task is to build and compare multiple classification models to predict whether a customer will churn. You will train different machine learning algorithms, evaluate their performance using appropriate classification metrics, and identify the most effective model for customer churn prediction. Finally, interpret the important factors influencing customer churn and communicate your findings through suitable visualizations and performance comparisons.

Tasks to Perform:

1. Use CustomerChurn_Preprocessed.csv file which you created in Ex2.
2. Split the cleaned dataset into Training (80%) and Testing (20%) datasets.

Model Building

3. Train a Logistic Regression classifier and predict the customer churn on the test dataset.
4. Train a Decision Tree Classifier and predict the customer churn on the test dataset.
5. Train a Random Forest Classifier and predict the customer churn on the test dataset.

Model Evaluation

6. Evaluate each model using Confusion Matrix, Accuracy, Precision, Recall, F1-Score.
7. Compare the performance of all three models using a single bar chart showing the evaluation metrics.
8. Identify the best-performing model based on the evaluation results.

Business Recommendation

9. Based on the best-performing model, provide three recommendations that the telecommunications company can implement to reduce customer churn.

Aim

To build Logistic Regression, Decision Tree and Random Forest classifiers for customer churn prediction, evaluate them with standard classification metrics and identify the best-performing model.

Procedure

1. Load CustomerChurn_Preprocessed.csv created in Ex2.
2. Split into features and target (Churn) and divide 80:20 into training and testing sets.
3. Train Logistic Regression, Decision Tree and Random Forest classifiers and predict on the test set.
4. Compute the confusion matrix, accuracy, precision, recall and F1-score for each model.
5. Compare all models in one grouped bar chart.
6. Select the best model from the metrics and give three business recommendations.

Code

```
#Mohammed Fayyad(241801165)
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (confusion_matrix, accuracy_score,
                             precision_score, recall_score, f1_score)

df = pd.read_csv("CustomerChurn_Preprocessed.csv")
X = df.drop(columns=["Churn", "customerID"], errors="ignore")
y = df["Churn"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(n_estimators=200, random_state=42),
}

rows = []
for name, m in models.items():
    m.fit(X_train, y_train)
    p = m.predict(X_test)
    print(name); print(confusion_matrix(y_test, p))
    rows.append([name, accuracy_score(y_test, p), precision_score(y_test, p),
                 recall_score(y_test, p), f1_score(y_test, p)])

res = pd.DataFrame(rows, columns=["Model", "Accuracy", "Precision", "Recall", "F1"])
print(res)
```

```
res.set_index("Model").plot(kind="bar", figsize=(9, 5)); plt.ylim(0, 1)
plt.title("Model Comparison"); plt.xticks(rotation=0); plt.show()
print("Best model:", res.sort_values("F1", ascending=False).iloc[0]["Model"])

# Feature importance of the Random Forest (for recommendations)
rf = models["Random Forest"]
pd.Series(rf.feature_importances_, index=X.columns).nlargest(10).plot(kind="barh");
plt.show()
```

Result

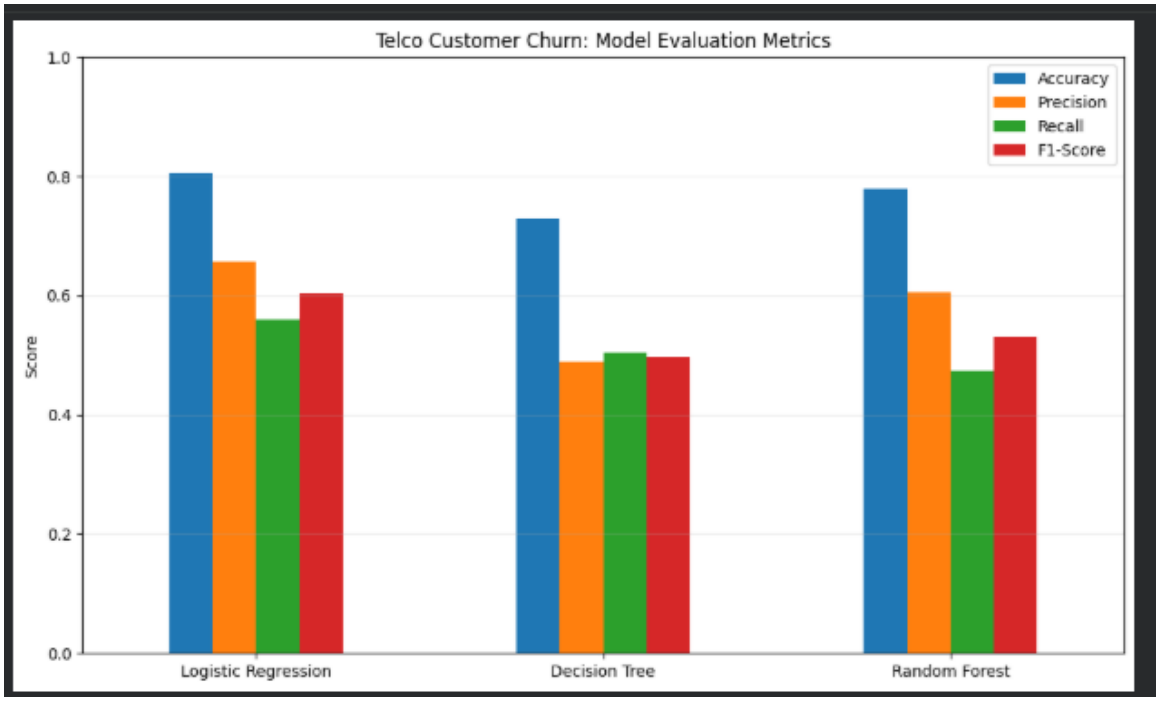
All three classifiers were trained and evaluated. The confusion matrix, accuracy, precision, recall and F1-score of each model were obtained, the bar chart compared the models, and the best model was selected on the evaluation metrics. Three retention recommendations were drawn from the most important features.

```
Logistic Regression Confusion Matrix [No Churn, Churn]:
[[926 109]
 [165 209]]

Decision Tree Confusion Matrix [No Churn, Churn]:
[[838 197]
 [185 189]]

Random Forest Confusion Matrix [No Churn, Churn]:
[[920 115]
 [197 177]]
```

	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8055	0.6572	0.5588	0.6040
Decision Tree	0.7289	0.4896	0.5053	0.4974
Random Forest	0.7786	0.6062	0.4733	0.5315



Top 10 Random Forest Features (Predictive Importance):

	Importance
num__TotalCharges	0.159271
num__tenure	0.139823
num__MonthlyCharges	0.137574
cat__Contract_Month-to-month	0.051714
cat__OnlineSecurity_No	0.032086
cat__PaymentMethod_Electronic check	0.030798
cat__InternetService_Fiber optic	0.027810
cat__TechSupport_No	0.027084
num__SeniorCitizen	0.019781
cat__gender_Female	0.017738